

JEE DROPOUT ANALYSIS

INTERNSHIP REPORT

Submitted By

ADVAITH G

(2301006)

NXTLOGIC SOFTWARE SOLUTIONS, COIMBATORE

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VATTAMALAIPALAYAM, N.G.G.O. COLONY POST,

COIMBATORE – 641 022.

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ANNA UNIVERSITY::CHENNAI 600 025

BONAFIDE CERTIFICATE

This is to certify that the Internship Training Report entitled “**JEE Dropout Analysis**” is a record of **ADVAITH G (2301006)** who carried out the internship training work under my supervision.

SIGNATURE

Dr.R. Kingsy Grace,
ACADEMIC COORDINATOR,
Professor,
Department of CSE,
Sri Ramakrishna Engineering College,
Vattamalaipalayam,
N.G.G.O.Colony Post,
Coimbatore - 641 022.

SIGNATURE

Dr.M.S. Geetha Devasena,
HEAD OF THE DEPARTMENT,
Professor and Head,
Department of CSE,
Sri Ramakrishna Engineering College,
Vattamalaipalayam,
N.G.G.O.Colony Post,
Coimbatore - 641 022.

Faculty In-charge

Signature of the Evaluator

Place: Coimbatore

Date:

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INTRODUCTION

I successfully completed a 14-day Internship at NxtLogic Software Solutions, where I gained practical exposure to Machine Learning concepts, industry-oriented workflows, and real-world project development. The internship helped me understand how data-driven solutions are designed, evaluated, and deployed to solve real-life problems. I chose Machine Learning and Data Analysis as my focus area due to my interest in predictive modeling. During the initial phase, I learned Python fundamentals for Machine Learning, followed by hands-on training in classical algorithms such as Linear and Logistic Regression, Decision Trees, Random Forest, KNN, SVM, and Boosting techniques. This enabled me to understand algorithm selection and performance evaluation.

As part of the internship, I developed a project titled “JEE Dropout Analysis”, aimed at predicting whether a student is likely to drop out after Class 12 based on academic and performance-related factors. The project involved data preprocessing, Exploratory Data Analysis (EDA), model training, and evaluation, and was implemented as a Flask-based web application for user interaction. In the final phase, I was introduced to Git and GitHub for version control, which provided insights into real-world code management and collaboration practices.

Overall, the internship at NxtLogic Software Solutions strengthened my foundation in Machine Learning and enhanced my confidence in building practical predictive systems

1. OVERVIEW OF THE ORGANIZATION

NextLogic Software Solutions, commonly known as *Nxtlogic*, is a technology and research-oriented company headquartered in Coimbatore, Tamil Nadu, India, established in 2009. It provides a wide range of IT services, digital solutions, academic support, and training programs to students, professionals, and businesses both in India and globally. NextLogic focuses on turning innovative ideas into powerful digital outcomes through a combination of cutting-edge technologies and practical expertise. The organization offers services such as web and mobile application development, artificial intelligence and machine learning solutions, cloud solutions, digital marketing, search engine optimization (SEO), and custom software solutions. In addition to technology services, NextLogic also provides PhD research assistance, including thesis and journal publication support, as well as internship and corporate training programs. Their internship training emphasizes hands-on experience, enabling learners to work on real-world projects in areas like Python, AI, data science, and full-stack development. NextLogic's mission is to empower learners, professionals, and organizations with practical skills, innovative tools, and professional support to help them succeed in a rapidly evolving digital environment. Its vision is to become a global leader in IT services, AI solutions, and academic support, driving digital growth and measurable impact for its stakeholders.

Office

No. 415/1, Vivekananda Rd,
Opposite of Zudio, Peranaidu
Layout, Ram Nagar, Gandhipuram,
Coimbatore – 641009, Tamil
Nadu, India

Type of the organization

IT Services, AI Solutions, and
Academic Support Provider

Years of Experience

16 Years (Founded 2009)

Nature of business

Provides IT solutions, software development, AI and machine learning services, digital marketing, research support, and industry-focused internship training

Business motto

“Turning digital visions into practical solutions through technology, research, and real-world training”

URL (website)

<https://www.nxtlogic.com/>

2. ABOUT MY PROJECT

I completed my internship at NextLogic in the Machine Learning and Development domain. As part of the internship, I worked on a project titled “JEE Dropout Analysis System.” The objective of this project is to identify students who are at risk of dropping out after Class 12 based on their academic performance and preparation-related factors. This system is designed to assist educational institutions and mentors in taking early preventive measures.

The project focuses on applying Machine Learning techniques to analyze student-related data and predict dropout likelihood. It enables users to input relevant academic and performance details and receive a prediction indicating whether the student is likely to continue or drop out. The system is intended to be simple and accessible, even for users without technical knowledge of Machine Learning.

The application includes multiple Machine Learning modules such as data preprocessing, Exploratory Data Analysis (EDA), and model training. Several classical Machine Learning algorithms including Logistic Regression, Decision Trees, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Boosting algorithms were implemented and evaluated. The best-performing model was selected based on performance metrics to ensure reliable predictions.

The technology stack used for this project includes HTML, CSS, and JavaScript for frontend development, Python and Flask for backend implementation, and Scikit-learn, Pandas, NumPy, and Matplotlib for Machine Learning and data analysis. The trained model and preprocessing components were saved and integrated into the web application to provide real-time predictions.

Overall, the JEE Dropout Analysis System is a Machine Learning-based web application that demonstrates how predictive analytics can be applied in the education domain. The project helped me gain practical experience in building end-to-end Machine Learning solutions, from data processing and model selection to deployment using a web framework.

2.1 Artificial Intelligence:

At its simplest form, artificial intelligence is a field, which combines computer science and robust datasets, to enable problem-solving. It also encompasses sub-fields of machine learning and deep learning, which are frequently mentioned in conjunction with artificial intelligence. These disciplines are comprised of AI algorithms which seek to create expert systems which make predictions or classifications based on input data.

Over the years, artificial intelligence has gone through many cycles of hype, but even to skeptics, the release of OpenAI's ChatGPT seems to mark a turning point. The last time generative AI loomed this large, the breakthroughs were in computer vision, but now the leap forward is in natural language processing. And it's not just language: Generative models can also learn the grammar of software code, molecules, natural images, and a variety of other data types. The applications for this technology are growing every day, and we're just starting to explore the possibilities. But as the hype around the use of AI in business takes off, conversations around ethics become critically important.

2.1.1 Machine Learning:

Machine Learning is a subset of artificial intelligence that enables systems to learn from data and improve their performance without being explicitly programmed. It focuses on developing algorithms that can identify patterns, make predictions, and support decision-making based on input data. By learning from historical data, machine learning models can adapt and improve their accuracy over time.

Machine learning powers many real-world applications that support automation and intelligent decision-making across industries. It is widely used in areas such as recommendation systems, spam detection, fraud detection, predictive analytics, medical diagnosis, and educational data analysis. By leveraging machine learning techniques, organizations can extract meaningful insights from large datasets and develop intelligent systems that assist humans in solving complex problems efficiently.

3. COMPONENTS

- User Interface Component
- Data Input Module
- Data Preprocessing Module
- Machine Learning Model Module
- Model Evaluation Module
- Backend (Flask) Module
- Model Serialization Module
- Result Display Module

3.1 User Interface Component:

The User Interface (UI) serves as the interaction point between the user and the system as shown Fig 3.1. It is designed to be simple and intuitive so that even users without technical knowledge can easily use the application.

- Provides input fields for entering student-related data
- Ensures proper validation of user inputs
- Displays prediction results clearly

The screenshot displays a dark-themed user interface. On the left, a 'Student Profile' form contains various input fields and dropdown menus for student data. On the right, a 'Model Performance' section features a bar chart comparing model performance across different categories.

Student Profile

Field	Field
JEE Main Score	JEE Advanced Score
Avg Mock Test Score	Class 12 Percentage
Attempt Count	Daily Study Hours
School Board	Coaching Institute
Family Income	Parent Education
Location Type	Peer Pressure Level
Mental Health Issues	Admission Taken

Model Performance

Enter details to see the prediction.

Model Performance Comparison

Category	Model A	Model B	Model C	Model D
Model A	0.85	0.75	0.65	0.55
Model B	0.95	0.95	0.95	0.95
Model C	0.95	0.95	0.95	0.95
Model D	0.95	0.95	0.95	0.95

Predict Future

Fig 3.1 User Interface Module

3.2 Data Input Module

This module is responsible for collecting student data required for prediction as shown in Fig 3.2. The inputs include academic performance indicators and preparation-related parameters.

- Accepts numerical and categorical inputs
- Performs basic validation before sending data to the backend
- Ensures consistency with the training dataset

JEE Main Score	JEE Advanced Score
33	55
Avg Mock Test Score	Class 12 Percentage
44	88.5
Attempt Count	Daily Study Hours
4	4
School Board	Coaching Institute
CBSE	FIITJEE
Family Income	Parent Education
Low	Graduate
Location Type	Select...
Rural	12th
Mental Health Issues	Graduate
No	PG
	Upto 10th
	ADMISSION TAKEN
	Yes

Fig 3.2 Data Input Module

3.3 Data Preprocessing Module

The dataset used for the JEE Dropout Analysis is a CSV as shown in Fig 3.3 file containing academic and preparation-related attributes such as JEE scores, mock test scores, school board, coaching type, study hours, and the target variable dropout. The dataset was examined to identify missing values, inconsistencies, and noise.

Categorical features were encoded into numerical form, missing and inconsistent values were handled, and extreme values were treated during Exploratory Data Analysis (EDA). Numerical features were scaled to ensure uniform contribution during model training, improving overall model reliability.

jee_main	jee_advance	mock_test	school_bor	class_12_f	attempt_c	coaching_i	daily_study	family_inc	parent_ed	location_ty	peer_pres	mental_h	admission	dropout
76.95	59.22	59.56	CBSE	70.09	1	FITJEE	5.4	Low	Upto 10th	Urban	Low	No	No	1
70.06	58.75	64.33	State	76	1	FITJEE	5.5	Mid	Upto 10th	Urban	Low	Yes	No	0
81.07	37.27	60.83	ICSE	64.36	1	FITJEE	7	Low	PG	Semi-Urba	Medium	Yes	No	1
93.39	60.72	69.33	ICSE	73.21	1	FITJEE	2.1	Low	12th	Semi-Urba	Medium	Yes	Yes	0
68.72	77.73	82.37	CBSE	89.02	1	Allen	6.3	Mid	Graduate	Semi-Urba	High	No	Yes	0
68.72	45.63	58.79	State	79.29	1	Local	4.5	Mid	Graduate	Semi-Urba	Low	No	Yes	0
94.11	82.78	80.01	CBSE	76.31	1	Local	3.7	Mid	Upto 10th	Semi-Urba	Medium	No	Yes	0
82.74	73.8	58.62	CBSE	77.15	1	None	2.2	High	12th	Urban	Low	No	Yes	0
65.43	54.25	57.63	ICSE	71.11	1	Local	5.7	Mid	Upto 10th	Semi-Urba	Low	No	Yes	0
79.6	63.74	77.82	State	94.95	2	None	2.7	Mid	12th	Rural	High	Yes	No	0
65.51	91.57	79.06	CBSE	66.89	1	None	5.5	Mid	Graduate	Urban	Low	No	Yes	0
65.48	77.74	78.22	ICSE	77.11	1	FITJEE	8.4	High	Upto 10th	Semi-Urba	Medium	Yes	Yes	0
75.39	66.14	52.21	CBSE	76.15	2	Allen	6.5	Mid	12th	Rural	Medium	No	Yes	0
45.21	62.2	62.41	ICSE	65.09	2	Allen	7.3	Low	Graduate	Urban	Medium	Yes	No	1
47.85	67.23	77.66	State	66.03	2	Local	3.1	Low	12th	Rural	Medium	Yes	No	1
64.13	57.81	68.5	CBSE	74.68	2	None	6	High	Graduate	Semi-Urba	Low	No	No	0
57.82	32.32	55.12	State	77.91	1	FITJEE	2.2	High	PG	Rural	Medium	No	Yes	0
76.4	98.67	73.5	State	78.29	1	None	4	High	PG	Urban	High	Yes	Yes	0
59.29	68.49	59.42	ICSE	52.28	1	FITJEE	8	High	12th	Urban	Low	No	No	0
52.23	72.37	89.53	CBSE	59.11	2	None	3.3	High	Upto 10th	Rural	Medium	Yes	No	0
92.52	65.07	86.54	CBSE	68.95	1	Local	6.7	Mid	12th	Rural	Medium	Yes	No	0
68.84	54.24	75.25	ICSE	69.38	1	Local	6.5	Low	PG	Urban	High	Yes	No	1
72.95	63.03	84.33	ICSE	63.01	2	None	6.6	Low	Upto 10th	Semi-Urba	Low	No	No	1
52.05	40.41	68.78	State	82.5	2	Allen	2	Mid	PG	Rural	High	Yes	Yes	0
64.38	100	77.19	CBSE	75.02	2	FITJEE	5.6	High	PG	Semi-Urba	High	Yes	Yes	0
73.95	45.25	85.73	CBSE	72.85	2	Allen	4.2	Mid	Upto 10th	Urban	Medium	No	Yes	0
55.89	49.49	70.95	ICSE	62.44	2	Allen	6.7	High	Upto 10th	Semi-Urba	Medium	Yes	Yes	0

Fig 3.3 Data Preprocessing Module

3.4 Machine Learning Model Module

The Machine Learning Model Module is the core component of the JEE Dropout Analysis System, responsible for analyzing processed student data and generating predictions. In this module, multiple classical Machine Learning algorithms such as Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM) were implemented.

Each model was trained using the preprocessed dataset and evaluated using appropriate performance metrics. Based on the evaluation results, the best-performing model was selected for final deployment. The selected model is used to predict whether a student is likely to drop out or continue after Class 12, ensuring reliable and accurate predictions.

3.5 Model Evaluation Module

The Model Evaluation Module was used to compare the performance of different Machine Learning algorithms implemented in the JEE Dropout Analysis System. Models such as Logistic Regression, Decision Tree, Random Forest, SVM, and KNN were evaluated using Accuracy, Precision, Recall, and F1-score to ensure reliable classification. From the performance comparison as shown in Fig 3.4, Decision Tree and Random Forest models showed the best results across all evaluation metrics. Logistic Regression exhibited lower recall and F1-score, while SVM and KNN produced moderate performance. The Decision Tree model was selected for deployment due to its high accuracy and consistent results.

The confusion matrix of the Decision Tree model as shown in Fig 3.5 indicates a high number of correct predictions with very few misclassifications, confirming the model’s effectiveness in distinguishing between dropout and non-dropout cases.

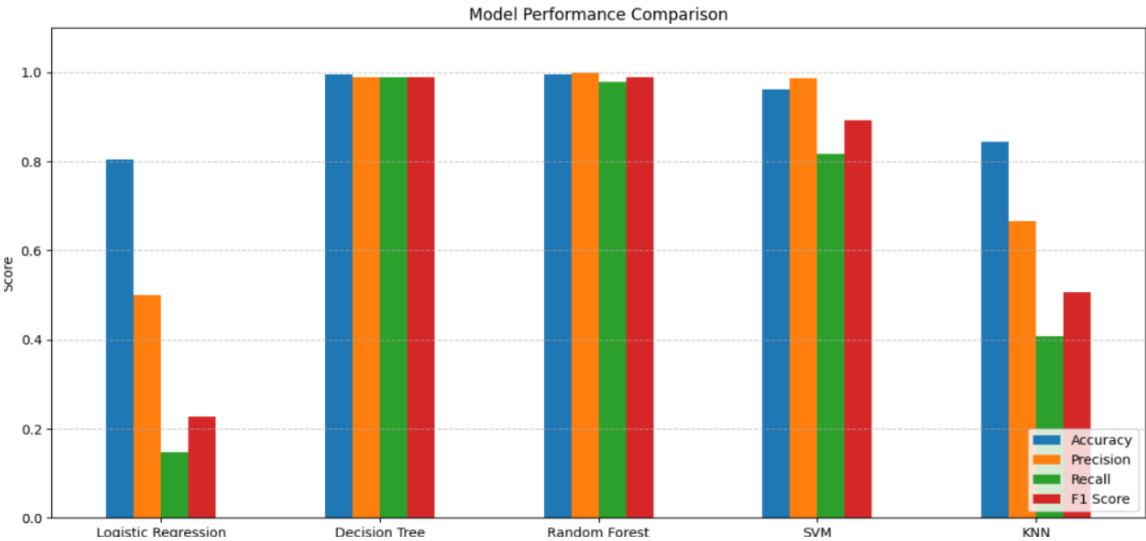


Fig 3.4 Machine Learning Models Comparison

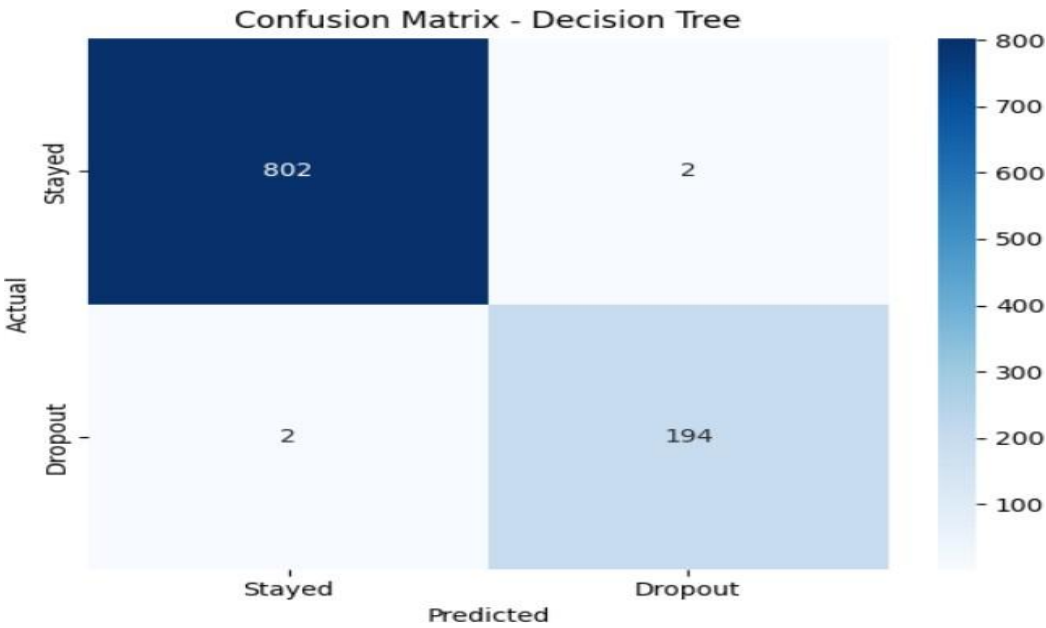


Fig 3.5 Confusion Matrix - Decision Tree Model

3.6 Backend (Flask) Module

The Backend Module of the JEE Dropout Analysis System is implemented using the Flask framework, which acts as a bridge between the user interface and the Machine Learning model. It is responsible for handling user requests, processing input data, and generating prediction results.

When a user submits student details through the frontend, the Flask backend receives the data through HTTP requests. The input data is then validated, preprocessed, and converted into a suitable format for the trained Machine Learning model. The backend loads the serialized model and preprocessing objects to perform real-time predictions.

After generating the prediction, the backend sends the result back to the frontend for display. This module ensures smooth communication between the frontend and the Machine Learning model, enabling efficient and reliable prediction without exposing the internal model logic to the user.

3.7 Model Serialization Module

The Model Serialization Module is used to store the trained Machine Learning model so that it can be reused without retraining. After the model was trained and evaluated, it was saved using the **joblib** library, which allows efficient storage of Machine Learning objects.

Along with the model, important preprocessing components such as encoders and scalers were also saved. This ensures that user input data is processed in the same way during prediction as it was during training. By using serialization, the system can load the trained model quickly whenever a prediction request is made. This approach improves the efficiency of the application, reduces computation time, and makes the system portable. It also allows the application to be deployed on different systems without the need to retrain the model again.

3.8 Result Display Module

The Result Display Module is responsible for presenting the prediction output to the user in a clear and understandable manner. After the Machine Learning model processes the input data, the prediction result is sent from the backend to the frontend for display. The system displays whether the student is predicted to **“Dropout”** or **“Not Dropout”** based on the model’s output. The result is shown in a user-friendly format so that even non-technical users can easily understand the prediction as shown in Fig 3.6. This module ensures that the outcome of the prediction is communicated effectively without exposing technical details.

By providing clear and meaningful results, this module helps users make informed decisions based on the model’s prediction.

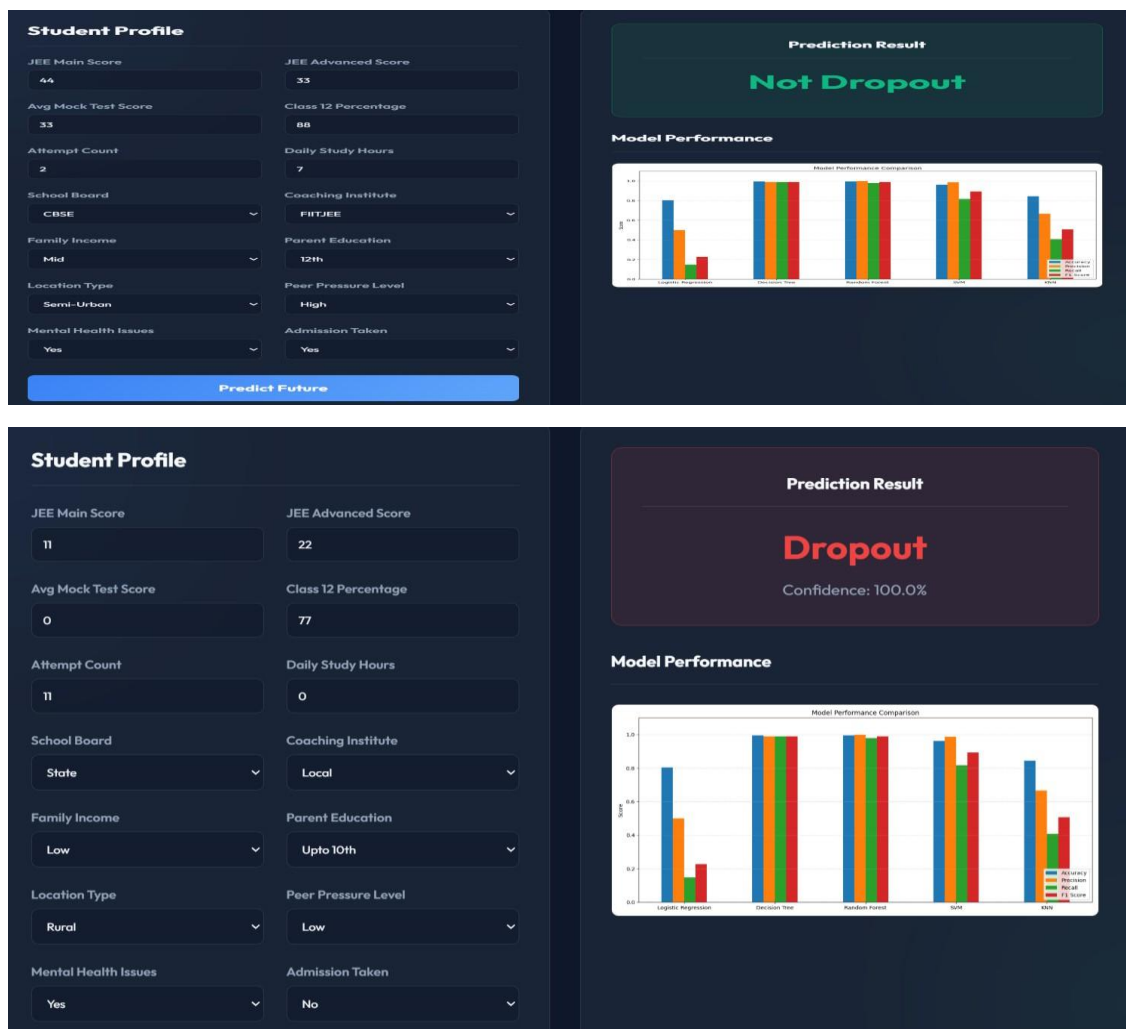


Fig 3.6 Results

4. CONCLUSION AND FUTURE SCOPE

From the analysis and implementation carried out during the internship, it is evident that Machine Learning techniques can be effectively applied in the education domain to address real-world problems such as student dropout prediction. The JEE Dropout Analysis System demonstrates how academic and preparation-related data can be converted into meaningful insights to support early intervention and informed decision-making.

The project involved implementing and evaluating multiple Machine Learning algorithms to identify the most suitable model for prediction. This helped in understanding the importance of data preprocessing, feature selection, and evaluation metrics in building reliable predictive systems. The successful deployment of the model as a web application highlights the practical usability of Machine Learning in real-world environments.

Future Scope

The system can be further enhanced by including additional parameters such as psychological factors, attendance patterns, and continuous assessment data. Advanced techniques such as ensemble learning and deep learning models can be explored to improve prediction accuracy. In the future, this system can be developed into a scalable educational analytics platform to support institutions in reducing student dropout rates and improving academic outcomes.

5. LESSONS I HAVE LEARNED

During the initial phase of my internship, I learned the fundamentals of Python programming required for Machine Learning, including data handling, conditional logic, and working with libraries such as Pandas and NumPy. This helped me strengthen my programming foundation and apply theoretical concepts in practical scenarios. I also learned how real-world datasets differ from academic examples and require careful analysis.

As the internship progressed, I gained hands-on experience with various Machine Learning algorithms such as Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, SVM, and Boosting techniques. I learned how to train models, evaluate their performance using appropriate metrics, and select the most suitable model based on problem requirements rather than accuracy alone.

I also learned the importance of data preprocessing and Exploratory Data Analysis (EDA) in Machine Learning projects. Handling missing values, encoding categorical features, and treating noisy data significantly improved model performance. This phase taught me that data quality plays a major role in building reliable predictive systems.

In addition to technical skills, I learned how to work in a professional learning environment, manage time effectively, and adapt to problem-solving under guidance. I was introduced to Git and GitHub, which helped me understand version control, code tracking, and collaborative development practices. Overall, the internship improved both my technical knowledge and professional confidence.

6. INTERACTION WITH CO-WORKERS

My interaction with the mentors and team members at **NxtLogic Software Solutions** was positive and encouraging throughout the internship period. The work environment was professional, friendly, and supportive, which made it easy for me to communicate and clarify doubts whenever required. My mentors were approachable and always willing to guide me during different stages of the project.

Whenever I faced challenges in understanding Machine Learning concepts or implementing algorithms, the mentors patiently explained the solutions and provided useful suggestions. I never felt hesitant to ask questions or share my ideas, as the team maintained an open and collaborative culture. Constructive feedback was given regularly, which helped me improve my technical understanding and problem-solving approach.

The guidance provided by the mentors helped me gain confidence in handling tasks independently while ensuring correctness. Their encouragement and continuous support motivated me to perform better and actively contribute to the project. Overall, the interaction with co-workers at NxtLogic was a valuable experience that enhanced my professional communication skills and teamwork abilities.

7. DIFFICULTIES ENCOUNTERED

The internship provided valuable real-world exposure, which helped me transition from academic learning to practical application. Overall, the working environment at NxtLogic Software Solutions was supportive, and the challenges faced were manageable due to the guidance provided by mentors and team members.

One of the main difficulties I encountered was understanding and implementing certain Machine Learning concepts during the initial stages, especially while working with real datasets. Handling data preprocessing, model evaluation, and interpreting results required continuous learning and practice. Additionally, managing time effectively while completing multiple tasks within the given schedule was initially challenging.

With regular guidance and support from my mentors, I gradually adapted to these challenges. Their suggestions and feedback helped me improve my understanding and complete tasks more efficiently. Over time, I became more confident in handling Machine Learning workflows and successfully completed the assigned tasks without major difficulties.

8. WEB REFERENCES

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